

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT)

PROFESSIONAL SUMMARY

Forward-deployed engineer with 2.5+ years shipping production AI services and integrations end-to-end - discovery, integration design, FastAPI/Docker, observability, post-deploy reliability - in regulated healthcare and oil & gas environments. Masters degree (Kent State). Track record: ~30% post-deploy defect reduction, ~35% retrieval precision gains, ~30% deployment-error reduction with audit-clean Jenkins CI/CD. Comfortable embedded with customers, debugging messy distributed systems, and translating field learnings into product improvements.

CORE COMPETENCIES

FastAPI • Docker • REST/GraphQL APIs • Webhooks • Auth/RBAC • Structured logging • Jenkins CI/CD •
Observability/Grafana • Customer-embedded delivery • RAG • Agentic LLM workflows • Tool calling • Vector search •
Postgres/SQL • Python • TypeScript • HIPAA-conscious governance

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Owned end-to-end delivery of healthcare AI workflows from data and prompts through evaluation, FastAPI/Flask APIs, and Docker-based deployment with structured logging; reduced post-deployment defects by ~30%.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Built **RAG systems** for clinical knowledge retrieval; improved retrieval precision ~35%, reduced irrelevant retrieval >30%, and pushed retrieval-grounded response alignment above 90%.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show prediction; improved recall 15-20% on high-risk categories.
- Built preprocessing pipelines for EHR extracts and support tickets; raised dataset reliability above 98% and cut downstream model instability by ~40%.
- Maintained HIPAA-conscious data governance, audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40% via structured release validation and rollback checkpoints.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; reduced manual review time by 60-70%.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension built solo (Manifest V3, content scripts, service workers, captureVisibleTab). Integrates four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline for tall webtoon panels with viewport-slice screenshots, coordinate remapping, and dedup. Seven-day translation cache, per-domain selector configs for 29 sites, narrowed host permissions, multi-provider payload handling (WebP cloud, JPEG Ollama).

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization to Validation to Consistency to Duplicate Detection to Fraud Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Intermediate structured prompt transformation layers raised contextual alignment ~30% and first-pass image success rate ~25-30% over naive direct-prompt orchestration.

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded 80-120 active users in phase one; cart, reviews, and community blogging features to drive repeat engagement.

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with a unified YOLOv8 detection-and-classification model; reduced inference latency ~30%. LabelImg annotations, augmentation for lighting/head-pose robustness, sliding-window confidence aggregation to suppress blink-driven false positives (~25% reduction), adaptive frame skipping, and NMS tuning for stable real-time operation via OpenCV.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

F-1 OPT - authorized to work in the United States; future sponsorship required

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems